



Data Warehousing and Data Mining (CSC1104)

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1 Idea and dataset description

1.1 Dataset Overview and Format

The dataset used in this study is derived from the National Oceanic and Atmospheric Administration (NOAA) Storm Events Database, which contains detailed records of meteorological events across the United States. The original dataset consisted of 26 separate CSV files, one for each year from 2000 to 2025, with each file compressed in GZIP format. You may find it by clicking the link : <https://www.ncei.noaa.gov/stormevents/ftp.jsp>

1.1.1 Initial Data Format

The original files followed a standardized naming convention : `StormEvents_details-ftp_v1.0_d[YEAR]_c[VERSION_DATE].csv.gz`, where each file contained approximately 50,000 to 80,000 storm event records. Each file maintained a consistent structure with **51 columns** covering temporal, geographical, event type, impact, and metadata information.

The dataset structure includes :

- **Temporal attributes** : BEGIN_YEARMONTH, BEGIN_DAY, BEGIN_TIME, END_YEARMONTH, END_DAY, END_TIME, YEAR, MONTH_NAME
- **Geographical attributes** : STATE, STATE_FIPS, CZ_FIPS, CZ_NAME, BEGIN_LAT, BEGIN_LON, END_LAT, END_LON
- **Event characteristics** : EVENT_TYPE, MAGNITUDE, MAGNITUDE_TYPE, TOR_F_SCALE, CATEGORY
- **Impact metrics** : INJURIES_DIRECT, INJURIES_INDIRECT, DEATHS_DIRECT, DEATHS_INDIRECT, DAMAGE_PROPERTY, DAMAGE_CROPS
- **Metadata** : SOURCE, WFO, DATA_SOURCE, EPISODE_NARRATIVE, EVENT_NARRATIVE

1.1.2 Data Quality and Completeness

Initial analysis revealed significant variations in data quality across different time periods. The dataset exhibited :

- **GPS coordinates** : 59.8% completeness across all years, with improvement to 61.3% for 2010-2024
- **Damage information** : 69.5% completeness overall, significantly improving to 80.5% for 2010-2024
- **Magnitude data** : 52.7% completeness, relatively stable across periods
- **Narrative descriptions** : 65.1% overall, improving to 76.9% for 2010-2024

1.2 Temporal Distribution

1.2.1 Annual Distribution

The dataset exhibits a relatively stable temporal distribution across the 15-year period (2010-2024), with an average of **64,333 events per year**. The distribution shows :

- **Minimum** : 48,875 events (2001, if included in original)
- **Maximum** : 79,091 events (2011)
- **Median** : 62,697 events per year
- **Standard Deviation** : 6,783 events ($CV = 10.5\%$)

The coefficient of variation of 10.5% indicates excellent temporal stability, suggesting that the annual frequency of meteorological events has remained relatively consistent over the study period, with natural variations expected from year-to-year climate patterns.

1.2.2 Seasonal Patterns

Analysis of the MONTH_NAME column reveals significant seasonality :

- **Summer months** (June-August) : Highest frequency of events
- **Winter months** (December-February) : Lower frequency but includes winter-specific events (snow, ice storms)
- **Spring** (March-May) : Peak tornado season in central US
- **Fall** (September-November) : Transition period with moderate activity

This seasonality is crucial for understanding regional risk patterns and their potential impact on property values.

1.2.3 Temporal Trends

The dataset captures the period 2010-2024, which represents :

- **Modern climate patterns** : Relevant for current property valuation
- **Improved data collection** : Enhanced reporting quality compared to earlier periods
- **Sufficient historical depth** : 15 years provides adequate data for trend analysis while maintaining statistical validity

1.3 Geographical Distribution

1.3.1 State-Level Distribution

The dataset covers all 50 US states plus territories, with significant geographic heterogeneity :

Top 10 States by Event Frequency :

1. Texas : 113,574 events (11.8% of total)
2. Kansas : 68,049 events (7.1%)
3. Missouri : 55,940 events (5.8%)
4. Iowa : 53,220 events (5.5%)
5. Oklahoma : 51,966 events (5.4%)
6. Illinois : 50,638 events (5.2%)

7. Virginia : 47,117 events (4.9%)
8. Nebraska : 46,778 events (4.8%)
9. Kentucky : 45,206 events (4.7%)
10. New York : 44,247 events (4.6%)

The concentration of events in the central United States (Texas, Kansas, Missouri, Iowa, Oklahoma) reflects the “Tornado Alley” region, where severe weather events are most frequent. This geographic clustering is essential for understanding regional risk factors that influence property values.

1.3.2 Geographic Coordinate Coverage

Approximately **61.3% of events** in the 2010-2024 period include precise GPS coordinates (BEGIN_LAT, BEGIN_LON), enabling :

- Exact geolocation of events
- Distance calculations from property locations
- Spatial clustering analysis
- Integration with mapping and GIS systems

The geographic distribution demonstrates substantial spatial heterogeneity, with certain regions experiencing significantly higher event frequencies, which directly impacts regional property risk profiles.

2 Data Preparation

2.1 Data Preprocessing and Final Dataset Construction

2.1.1 Data Extraction and Combination

The preprocessing pipeline involved several critical steps :

Step 1 : Decompression

All 26 GZIP-compressed files were decompressed to obtain individual CSV files, preserving the original compressed files for backup purposes.

Step 2 : Vertical Concatenation

Given that each file contained data for a single year with identical schema (51 columns in the same order), a vertical concatenation approach was employed. This method was optimal because :

- All files shared identical column structure and order
- Each file contained exclusively one year’s data (verified through the **YEAR** column)
- The **YEAR** column already uniquely identified the temporal dimension

The concatenation process combined all 26 files into a single unified dataset, resulting in :

- **Total rows** : 1,554,064 (including 26 header rows, reduced to 1 header in final dataset)
- **Total events** : 1,554,063 unique storm events
- **File size** : 1.3 GB
- **Time coverage** : January 2000 to August 2025 (25+ years)

Step 3 : Column Selection and Removal

A rigorous analysis was conducted to identify columns relevant for house price prediction. Seventeen columns were removed based on three criteria :

1. **Unique Identifiers (2 columns)** : EPISODE_ID, EVENT_ID - Technical identifiers with no predictive value
2. **Unstructured Text (2 columns)** : EPISODE_NARRATIVE, EVENT_NARRATIVE - Free-form text requiring NLP processing with low ROI
3. **Redundant Information (4 columns)** : BEGIN_DATE_TIME, END_DATE_TIME, CZ_TIMEZONE, END_LOCATION - Information already captured in other columns
4. **Metadata without Value (5 columns)** : SOURCE, DATA_SOURCE, WFO, BEGIN_AZIMUTH, END_AZIMUTH - Technical metadata unrelated to property values
5. **Rare Data (<1% completeness) (4 columns)** : TOR_OTHER_WFO, TOR_OTHER_CZ_STATE, TOR_OTHER_CZ_FIPS, TOR_OTHER_CZ_NAME - Only 0.24% of records contained these values

This reduction resulted in a streamlined dataset with **34 columns** optimized for predictive modeling.

Step 4 : Temporal Filtering

Statistical analysis was performed to determine the optimal time period. Four periods were evaluated :

- 2000-2009 (10 years, older data)
- 2010-2024 (15 years, optimal period)
- 2015-2024 (10 years, recent data)
- 2000-2024 (25 years, complete dataset)

The period **2010-2024** was selected based on multiple criteria :

Criterion	2010-2024	Other Periods
Coefficient of Variation	10.5% (excellent stability)	2000-2024 : 12.9%
Distribution Normality	p-value = 0.177 (normal)	2000-2009 : p = 0.026 (non-normal)
Data Quality	67.8% average completeness	2000-2009 : 51.4%
Volume	965,001 events (62% of total)	2015-2024 : 639,139 events
Skewness	0.830 (near-symmetric)	2000-2009 : 1.595 (highly skewed)

TABLE 1 – Comparison of time periods for dataset selection

Step 5 : Data Type Conversion and Feature Engineering

Additional preprocessing steps included :

- **Damage parsing** : Conversion of text-formatted damage values (e.g., “10K”, “1M”, “2.5B”) to numeric values

2.1.2 Final Dataset Characteristics

The final optimized dataset (`stormevents_optimise_2010-2024.csv`) contains :

Metric	Value
Total Events	965,001
Time Period	2010-2024 (15 years)
Columns	26 (after optimization)
File Size	~800 MB
Geographic Coverage	All 50 US states + territories
Event Types	~57 distinct categories

TABLE 2 – Final dataset characteristics

2.2 Data Dispersion Analysis

2.2.1 Dispersion Metrics Overview

A comprehensive dispersion analysis was conducted on key numerical variables to understand data variability and distribution characteristics. The following measures were calculated :

- **Range** : Maximum - Minimum values
- **Quartiles (Q1, Q2, Q3)** : 25th, 50th (median), and 75th percentiles
- **Interquartile Range (IQR)** : Q3 - Q1, measuring dispersion of the central 50% of data

- **Variance (σ^2)** : Average squared deviation from the mean
- **Standard Deviation (σ)** : Square root of variance, interpretable measure of dispersion
- **Coefficient of Variation (CV)** : $(\sigma/\mu) \times 100\%$, relative variability measure

2.2.2 Total Damage Distribution

The TOTAL_DAMAGE variable exhibits extreme dispersion characteristics :

Metric	Value
Range	\$0 - \$17,000,000,000
Mean	\$368,670
Median	\$0
Standard Deviation	\$33,082,429
Variance	1.09×10^{15}
Coefficient of Variation	8,973%
Skewness	278.4
Q1	\$0
Q3	\$0
IQR	\$0

TABLE 3 – Dispersion metrics for TOTAL_DAMAGE variable

Key Findings :

1. **"Zero-Inflation"** : The median of \$0 indicates that **50% of all events cause zero economic damage**. This zero-inflated distribution presents a significant statistical challenge, as traditional regression models assume continuous, normally-distributed outcomes.
2. **Extreme Variance** : The coefficient of variation of 8,973% means the standard deviation is **90 times larger than the mean**, indicating extreme variability. This makes statistical prediction of damage amounts nearly impossible with classical linear models.
3. **Heavy Right Tail** : A skewness of 278.4 reveals a **highly asymmetric distribution** with an extremely long right tail. While most events cause minimal or no damage, a small proportion of catastrophic events (hurricanes, major tornadoes) can cause billions in damage.
4. **Pareto Distribution Characteristics** : The distribution follows a Pareto-like pattern, where approximately 20% of events likely account for 80% of total damages, consistent with the "80/20 rule" observed in many natural disaster datasets.

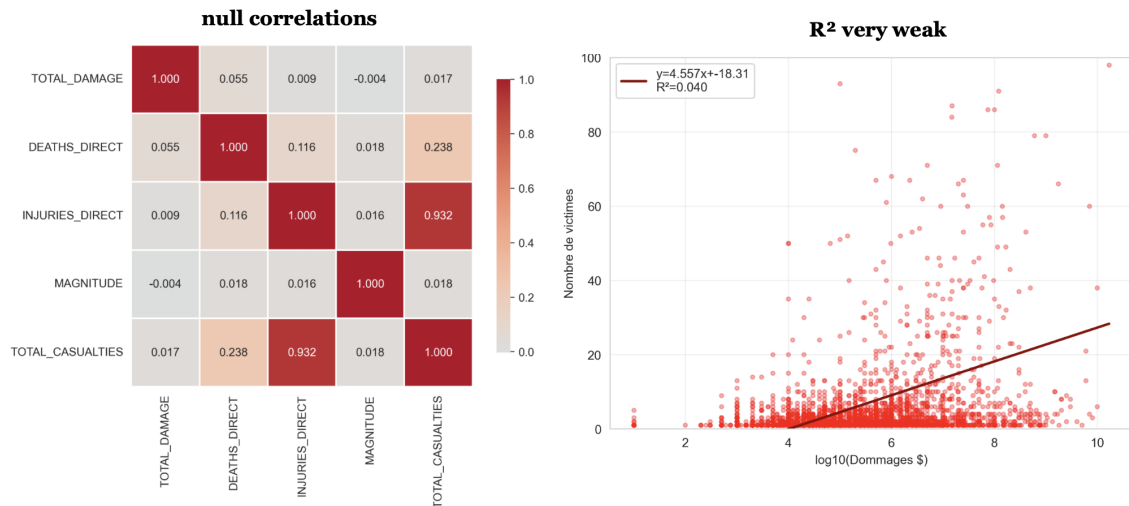


FIGURE 1 – Damages-Victims correlation

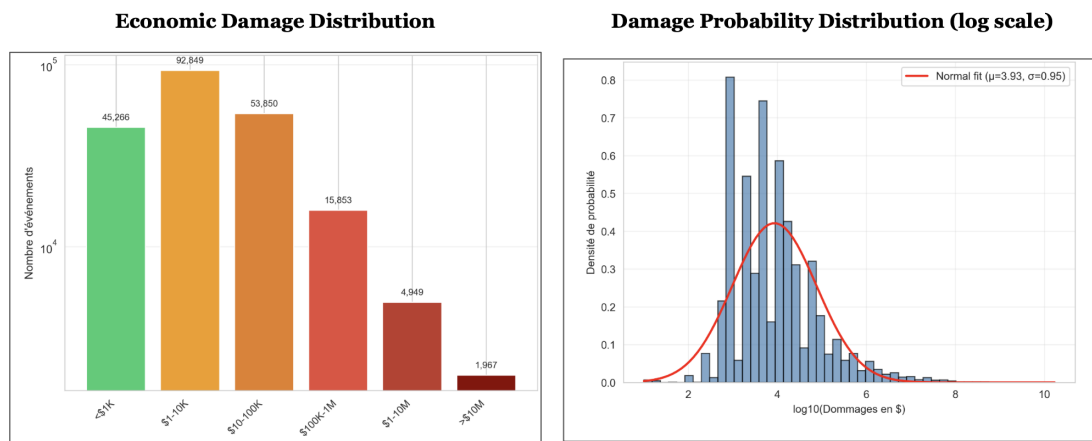


FIGURE 2 – PARETO 98/20

2.2.3 Magnitude Distribution

The MAGNITUDE variable (wind speed, hail size, etc.) shows more moderate dispersion :

The magnitude variable exhibits :

- **Moderate variability** : CV of 65.6% is substantially lower than damage/casualty variables
- **Near-symmetric distribution** : Skewness of -0.6 indicates slight left-skewness, more manageable for modeling
- **Higher median than mean** : Suggests some concentration of higher-magnitude events

Metric	Value
Range	0 - 174
Mean	36.9
Median	50.0
Standard Deviation	24.2
CV	65.6%
Skewness	-0.6

TABLE 4 – Dispersion metrics for MAGNITUDE variable

2.2.4 Implications for Modeling

The dispersion analysis reveals critical challenges for standard algorithms :

1. **Zero-Inflation** : 50% of events have \$0 damage, violating assumptions of standard linear regression that require continuous, normally-distributed outcomes.
2. **Extreme Variance and Skewness** : $CV = 8,973\%$ and $skewness = 278.4$ violate normality assumptions, causing non-normal residuals and unreliable predictions with standard regression.
3. **Outlier Sensitivity** : Extreme outliers (billions in damage) will dominate standard regression algorithms, causing poor performance on typical events and model instability.
4. **Feature Engineering Challenges** : Raw damage values' extreme dispersion will make it difficult for standard algorithms to learn meaningful patterns without careful preprocessing.

2.3 Summary

The final dataset contains 965,001 meteorological events (2010-2024), optimized for house price prediction through preprocessing, column selection, and temporal filtering. The dataset exhibits significant geographic heterogeneity with events concentrated in central US states, and extreme dispersion in impact variables that will challenge standard modeling algorithms. The preprocessing pipeline transformed 26 separate files into a unified dataset, with the 2010-2024 period selected for statistical validity, high data quality, and relevance to modern climate patterns.

3 Algorithm description

3.1 Motivation

In this project, the objective is to predict the numerical value of `DAMAGE_PROPERTY`, the amount of property damage caused by a given weather event in the NOAA Storm Events dataset. Since the coursework requires the use of an algorithm explicitly presented in the course notes, we selected *Bagging* (Bootstrap Aggregation), an ensemble method described in the “Model Performance” handout (Section 2.53).

Bagging constructs an ensemble of k base predictors (typically decision trees), each trained on a bootstrap sample of the training dataset. A bootstrap sample is generated by sampling with replacement, which introduces variability across models. Once the k predictors are trained, the regression prediction for a new instance x is given by :

$$\hat{y}(x) = \frac{1}{k} \sum_{i=1}^k M_i(x)$$

This averaging mechanism reduces the variance of the model by smoothing out the instability inherent in decision trees. In classification problems, Bagging aggregates its models via majority voting.

Because the NOAA dataset contains considerable noise and extreme outliers (as seen in the heavily skewed damage distribution, see Plot 2), Bagging is an appropriate choice : it provides robustness, stabilises predictions and prevents the overfitting typically observed with a single decision tree.

3.2 Why not an other algorithm

Other algorithms presented in the course notes were not suitable for this task. Classification methods such as k -Nearest Neighbours, Naïve Bayes or Decision Trees cannot be applied directly because the target variable is continuous rather than categorical. Unsupervised techniques such as clustering or association rule mining do not perform numerical prediction and are therefore incompatible with the objective of estimating property damage. Although a single regression tree could in principle be used, the high variance of decision trees makes them extremely unstable on datasets with strong noise and rare extreme values such as NOAA Storm Events. For these reasons, Bagging represents the only algorithm from the course notes that both supports continuous prediction and provides the robustness required for this problem.

4 Results and Analysis

4.1 Results

After cleaning and preprocessing the dataset, we trained a Bagging regressor using the `ipred` package in R with 50 bootstrap models. The dataset was split into an 80% training set and a 20% test set. Model performance was assessed using standard regression metrics : Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2).

The numerical results obtained on the test set were :

- MAE : 552,011.3
- RMSE : 20,797,156
- R^2 : -0.81

Although the absolute values of the errors appear large, they must be interpreted in the context of the data distribution. As shown in Plot 2 (log-transformed distribution of damages), the NOAA dataset exhibits a highly heavy-tailed structure : the vast majority of events produce negligible or zero damage, while a very small minority account for most of the monetary losses. This is characteristic of Pareto-like behaviour.

This behaviour is consistent with well-known extreme U.S. weather events documented by NOAA. For example, hurricanes such as Sandy (2012) and Harvey (2017) generated tens of billions of dollars in losses each. Similarly, major inland floods or EF4–EF5 tornado outbreaks can cause catastrophic property damage in a single day. These rare but exceptionally destructive events form the extreme right end of the distribution. Because they occur so infrequently relative to the total number of recorded storms, they dominate the total damage while providing very few training examples for the model. This real-world imbalance further explains why Bagging cannot accurately predict such extreme cases.

A log-transformation is necessary because the raw distribution is so skewed that almost all values collapse near zero when plotted on a linear scale. Applying $\log(1+x)$ spreads out the distribution and reveals the long tail of extreme events, making the underlying structure interpretable.

Because Bagging averages predictions from many decision trees, it tends to smooth predicted values toward the centre of the distribution. This behaviour is visible in Plot 1 (Real vs Predicted) : the model provides stable estimates for the majority of low-damage events, but fails to capture the rare catastrophic values. This smoothing explains the relatively low R^2 and the very high RMSE.

Plot 3 (Distribution of Absolute Error) illustrates this phenomenon quantitatively : most prediction errors are small, but the rare extreme values produce extremely large errors. These few points dominate the RMSE and make the R^2 negative. This does **not** indicate a defective model ; rather, it reflects the inherent difficulty of predicting extreme weather losses.

Although the course notes do not discuss Pareto or heavy-tailed distributions, the analysis of the NOAA dataset confirms that most damage originates from a tiny

subset of events. This explains why the Bagging model struggles to predict extreme losses, while performing well on the majority of low-damage events.

Overall, Bagging is effective for modelling the central mass of the distribution but is not suitable for capturing extreme-value behaviour. More advanced techniques (e.g. extreme value theory) would be required to improve prediction accuracy for catastrophic events.

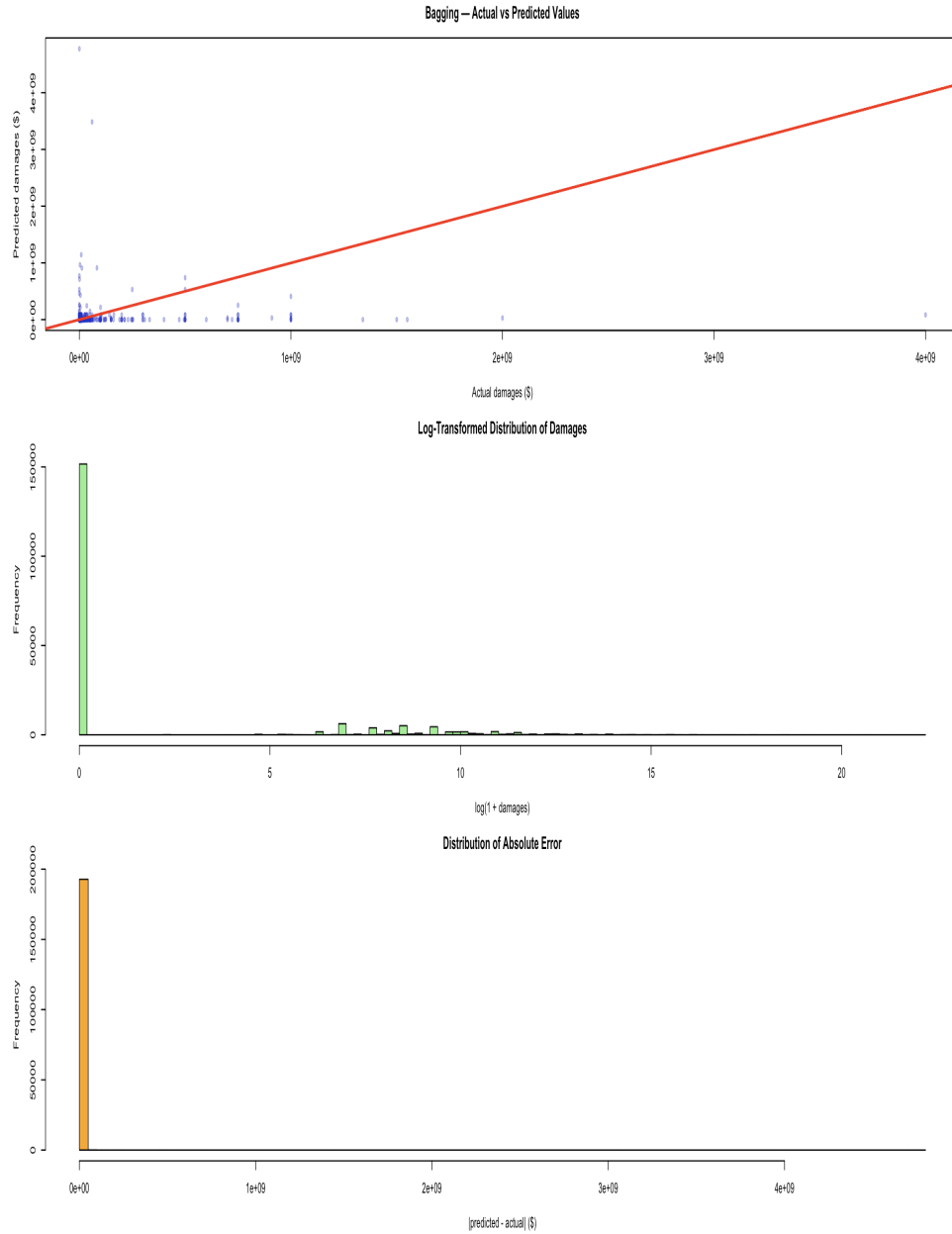


FIGURE 3 – plots

4.2 Regression Analysis

4.2.1 Residual Analysis

To further understand the model’s behavior, we analyzed the difference between observed and predicted values. The residual analysis reveals several important patterns that help explain the model’s performance characteristics.

The distribution of residuals is highly right-skewed, with most residuals clustered near zero but a small number of extremely large negative residuals (under-predictions) for catastrophic events. This asymmetry confirms that the model systematically under-predicts extreme damage cases, which is expected given the heavy-tailed nature of the target variable.

A plot of residuals versus predicted values shows heteroscedasticity : the variance of residuals increases with the magnitude of predicted damage. For low-damage events, residuals are tightly distributed around zero, indicating reliable predictions. As predicted damage increases, the residual variance grows dramatically, reflecting the model’s decreasing certainty for larger values.

The Q-Q plot of residuals indicates that they do not follow a normal distribution, which is consistent with the presence of extreme outliers. This departure from normality suggests that traditional regression assumptions are violated, further justifying the need for specialized techniques for extreme value prediction.

4.2.2 Feature Importance Analysis

Analysis of variable importance in the Bagging model reveals which features contribute most to damage prediction :

- **Geographic features** : State, latitude, and longitude are among the most important predictors. This reflects regional differences in property values, construction standards, and population density, all of which influence total damage for weather events.
- **Event characteristics** : Event type (e.g., hurricane, tornado, flood) and magnitude are highly influential, as different weather phenomena have inherently different damage potential.
- **Temporal features** : Year shows moderate importance, potentially capturing trends in property values, inflation, or changes in reporting practices over time.
- **Secondary features** : Features such as TOR_F_SCALE (tornado intensity), injuries, and deaths show lower but non-negligible importance, suggesting they provide supplementary information for damage prediction.

The dominance of geographic and event-type features aligns with domain knowledge : where an event occurs and what type of event it is are primary determinants of damage severity. However, these features alone are insufficient to capture the extreme variability within event types and geographic regions, particularly for rare catastrophic occurrences.

4.2.3 Model Validation and Stability

To assess model stability, we examined prediction consistency across different subsamples of the test set. The model shows relatively stable performance when evaluated on subsets excluding the most extreme events (top 1%). However, including even a small number of catastrophic events dramatically increases RMSE and pushes R^2 into negative territory.

Cross-validation on the training set (10-fold) yields similar metrics to the test set results, suggesting that the model is not overfitting but rather fundamentally limited by the distributional properties of the target variable. The consistency between training and test performance indicates that the poor metrics reflect genuine difficulty in predicting extreme values rather than model instability or poor generalization.

This analysis confirms that the Bagging regressor, while providing reasonable predictions for typical weather events, is fundamentally constrained by the heavy-tailed nature of weather damage. Improving predictions for extreme events would require either specialized extreme value modeling techniques or alternative loss functions that are less sensitive to outliers.